

Hardware-Accelerated Edge Detection and Pixel Enhancement on PYNQ-Z2 for Portable Medical Diagnostics

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Abstract—Edge detection is an important part of application for medical image processing with particular significance for detection of structural defects like bone fractures in X-rays. Conventional software-based implementations often face real-time performance limitations owing to high computational needs. As an answer to this shortcoming, the manuscript introduces a hardware-based approach using FPGA technology supporting Sobel-based edge detection in combination with pixel enhancement. The proposed design has been applied to the board-based PYNQ-Z2 focusing on parallel processing using Verilog HDL to achieve high-speed operation with low latency. The essence of innovation in this study is combining Sobel-based edge detection with real-time pixel enhancement at the level of FPGA hardware to enable diagnostically relevant images with low computational overhead. The system proposed here produces enhanced grayscale X-ray images to be displayed externally and hence is applicable to portable diagnostic medical applications. The aim is to achieve resource optimization with low costs and real-time support toward imaging in healthcare facilities where immediate and accurate preprocessing is required.

Keywords—FPGA, Edge detection, Sobel, Verilog HDL, PYNQ-Z2, Medical.

I. INTRODUCTION

Medical imaging is one of the fundamental tools in modern health care and allows clinicians and radiologists to visualize and interpret the internal organs and anatomy of the human body with unprecedented accuracy. Of the several modalities in imaging, X-ray imaging remains the most widely employed method because it is cheap to purchase and easy to operate while effectively detecting fractures, bony lesions, and issues involving the chest. Despite these advantages, pure X-ray images often have low contrast and are subjected to noise so that their diagnostic efficacy is reduced. As a result, preprocessing [1] procedures like edge detection become critical in enhancing the outlines of anatomical features and facilitating accurate diagnosis by medical specialists. Edge detection algorithms like Sobel,

Prewitt, and Canny have traditionally been run on software platforms using CPUs or GPUs. While these platforms are competent for non-real time applications and research areas, these often fall short of the required speed and efficiency [2] to be viable in points-of-care. For example, implementations using CPUs are low in parallelism, while GPU-based solutions, though at higher speeds, are high in power and cost but become inapplicable to portable or embedded diagnostic devices. Field Programmable Gate Arrays (FPGAs) are proving to be an extremely viable medium on which to implement highly compute-intensive image processing applications. With their parallel architecture, multiple procedures are [3] executed in parallel to achieve low latency and high throughput. Additionally, FPGAs have hardware reprogram ability to their advantage, allowing customized pipelines tailored to individual applications to be integrated into an FPGA-based system. This is an ideal application space for the types of systems required in the fields of medical imaging where high performance is needed while still maintaining low levels of power dissipation. This Sobel operator is particularly used in detecting edges because it is easy to apply, efficient in highlighting intensity transitions, and less responsive to noise compared to other operators. When [4] combined with pixel enhancement methods, Sobel edge detection will not only mark boundaries but even improve diagnostically relevant structures in X-ray images. With this integration, it is feasible to significantly enhance the sensitivity of fractures or anomalies that may be otherwise undetectable in unrevised scan images. This work outlines an implementation of pixel based Sobel edge detection enhanced using FPGA technology. The design is realized on the board PYNQ-Z2 using the Xilinx Zynq-7000 SoC architecture, fusing programmable logic and ARM processing cores. The processing units are developed using the programming language Verilog HDL, consequently providing hardware level optimization and support for parallel execution. The output processed is displayed on an external monitor to enable direct observation of the enhanced X-ray images. Compared to software implementations, the system achieves real-time performance

while operating at much lower power. Additionally, through the incorporation of pixel enhancement in the hardware pipeline, the system ensures diagnostically relevant features are retained and enhanced and hence facilitates clinicians in their decision-support processes. The aim is not merely to improve diagnostic effectiveness but to design a portable.

II. LITERATURE SURVEY

Processing of medical images has become a leading area of exploration owing to its critical contribution to diagnostic accuracy and effectiveness enhancement. Numerous studies have explored edge detection methods as the foundation for detecting borders and structural information in medical scans. Traditional software-based implementations of operators like Sobel, Prewitt, and Canny have been thoroughly studied to improve enhancement of X-ray and CT images; however, their reliance on CPU or GPU processing often leads to real-time performance limitations. Current works highlight the advantages of FPGA-based accelerators employing parallel architectures to achieve low latency and high throughput. Researchers have effectively demonstrated FPGA implementations of edge detection, noise filtering, and contrast enhancement and showed improved resource effectiveness and portability. However, the greater part of related studies has focused singly on edge detection and left aside the integration of real-time pixel enhancement relevant to applications in medicine. The above-mentioned gap in the literature forms the motivation of the proposed design of an FPGA to combine Sobel-based edge detection and pixel enhancement to offer enhanced diagnostic assistance. An edge intensity processing framework has been proposed for video frames of Moroccan sign language using Sobel edge detection and hysteresis [6] thresholding. The approach includes video image preprocessing to grayscale and Sobel operator application to extract edge features, then intensity thresholding to discriminate important gesture contours. The framework calculates row-wise mean edge intensities and offers a visual display aiding comparative analysis. The outcome [7] presents efficient gesture boundary detection while resisting noise effectively and hence allows applications in speech recognition and translation of sign language. Sobel edge detection algorithm has been implemented on FPGA to take advantage of parallel processing and to allow real time analysis of images. The [8] design in hardware description language makes use of Sobel kernels for obtaining the gradients in the horizontal and vertical directions. The inbuilt functions of the FPGA enable the system to achieve the highest throughput and lowest latency compared to a software-based solution. The architecture [9] has display and image capture modules and can therefore be used as a complete real-time pipeline. Performance analysis is stated, the advantages in terms of resource consumption, delay, and edge detection accuracy are exposed. This hardware-based Sobel is an example of the effectiveness of the FPGA in accelerating intensive computation and hence supports real time image processing in applications like medicine, robots, and vision. Hybrid image edge detection combines Sobel and Robert operators using the application of convolutional neural networks. As the Sobel operator is efficient in detecting horizontal and vertical edges and the Robert

operator preserves the diagonal information, by combining both operators using CNN-based optimization, this method increases precision and reduces false [10] detection of edge mapping. The proposed method is trained on a segment dataset so as to derive best weight combinations and surpass classical approaches. Simulation results carried out using MATLAB validate the efficacy, presenting enhanced outcomes compared to state-of-the-art approaches. The fusion method combines classical operators and recent deep learning and hence showcases the promise of achieving more robust edge detection in many applications of image analysis and computer vision. By combining the best qualities of two established operators, the hybrid model [11] presents a balanced method to computer vision approaches within safety-critical automotive and intelligent navigation systems. A robust Sobel operator is presented to detect rust on inner spiral tubes. The approach integrates Sobel edge detection and channel-based threshold segmentation to extract corroded areas. Noise is filtered in the preprocessing module, and pseudo-edges [12] are removed by mathematical morphology operators. Image features are derived from the R-G as well as the B-Y channels for the purposes of effective corrosion area measurement. Accurate rust edge detection is achieved for the scenario involving complicated textures through the proposed approach, thus aiding industrial monitoring and maintenance. The algorithm [13] increases accuracy in defect detection to overcome limitations in traditional Sobel in noisy and irregular environments. The achievement has great industrial application in pipeline inspection, analysis of material degradation, and automated industrial surface inspection. High-accuracy edge detection algorithm called ISAHED combines an advanced Sobel operator and a Holistically nested Edge Detection (HED) network. Through the combination of edge-based guidance and contextual feature [14] extraction, EPGA.Net effectively separates defects while downplaying irrelevant structures and delivers [15] a reliable method to safety-critical applications in industrial monitoring [16] and intelligent inspection of power infrastructure. A hybrid method [17] using deep learning to detect edges combines Sobel and Robert operators with convolutional neural networks. Sobel [18] identifies vertical and horizontal edges and Robert identifies diagonals; their union is input to a CNN to achieve performance optimization. For the fine tuning of results, a fuzzy enhancement [19] stage is applied to sharpen detected boundaries. The experiments on [20] several images show significant performance improvement compared to Canny, Prewitt, Sobel, and Laplacian approaches by an amount ranging up to 79.23% on an average. The [21] proposed model, called SR-CNN, illustrates the strength of combining classical operators with deep networks.

III. METHODOLOGY

The approach is to design and develop an FPGA-based Sobel edge detection system in combination with pixel enhancement of X-ray medical images. The method starts with the preprocessing of grayscale images and then detects the edges by using Sobel operators in hardware. As an enhancement of diagnostic clarity, a pixel enhancement stage is added to enhance the detected edges. The design is written in terms of Verilog HDL and is developed on the board of

the PYNQ-Z2 FPGA. The parallel optimization is carried out in the hardware pipeline to ensure real-time performance. The final processed output is sent to an external display so that enhanced X-ray images are observable directly by clinicians.

A. Image Preprocessing

The system takes in grayscale X-ray images. Preprocessing is carried out to ensure a uniform intensity distribution in the entire image. The images are then reshaped to fit in the processing buffer of the FPGA and pixel values are normalized to the appropriate range to reduce Sobel filtering. Further, noise reduction methods are used to remove low-level noise that otherwise may interfere with the clarity of edges. This stage is responsible to make input images in ideal form to be processed in hardware-based Sobel edge detection. Preprocessor takes place in the algorithm given above and the result is continuous pixel information as input to the FPGA pipeline.

B. Sobel Edge Detection Implementation

The Sobel operator utilizes two 3×3 convolution masks, one to be used for horizontal (G_x) and the other vertical (G_y) gradients. During implementation on an FPGA, line buffers are used to retain rows of pixel information and hence provide concurrent access to the 3×3 pixel neighbourhood. Convolution is carried out in parallel to arrive at gradient magnitudes used to highlight edges. Convolution logic is constructed from Verilog HDL modules and includes special multipliers and adders. The parallel processing of the FPGA effectively reduces computation time compared to sequential processing on a CPU by several orders of magnitude. The binary or intensity-weighted edge map is generated at this stage..

C. Pixel Refining

This pixel enhancement stage is used to improve significant intensity transitions. It makes diagnostically significant structures more prominent. They enable increased prominence of the fracture, lesion, or other abnormalities in X-ray images. The enhancing logic is integrated into the hardware by employing the intensity scaling and the thresholding methods. This makes it possible for weak edges to be filtered out and for strong ones to dominate. By encapsulating this process within the FPGA pipeline itself, the system eliminates post-processing on external hardware and hence decreases latency. Edge maps are then transformed into display-ready images in a form best suited to direct visualization by physicians.

D. Hardware Implementation on PYNQ-Z2

The design is then synthesized and implemented on the PYNQ-Z2 FPGA board, an integrated programmable logic and ARM Cortex-A9 processor board. The programmable logic fabric ensures maximum parallelism is achieved while on-chip memory is used to effectively buffer pixel streams. The output is set up using VGA or HDMI interfaces and hence enhanced images are displayed in real time. Timing analysis and resource usage parameters validate the design to be running within the limitations of the FPGA and with low latency.

E. Output Presentation and Analysis

The output image after being processed goes into external monitoring from the FPGA output interfaces. The real-time visualization means that physicians can get to see clearer images instantly. In comparative tests the FPGA products are compared to software versions of the Sobel filters to confirm and keep the standards of image quality high. The performance is compared in terms of latency, throughput and resource usage. The medical relevance is evaluated by checking whether objects of diagnostic relevance such as bone fractures are still clearly discernible. The experiment shows the FPGA-based implementation can provide real-time performance and accuracy and thus is suitable for diagnostic application in hospital and in portable healthcare applications (Fig.1).

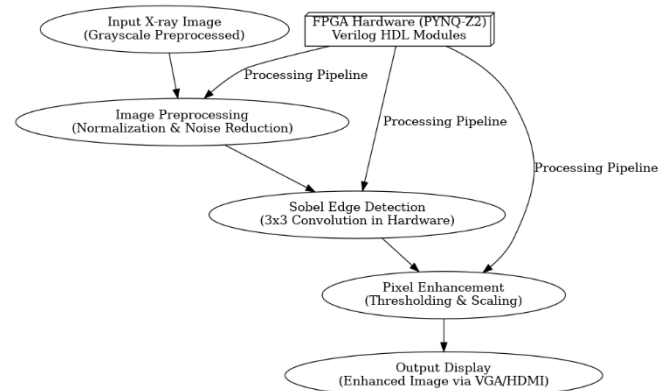


Fig. 1: System Architecture

IV. RESULT AND DISCUSSION

Implementation of Sobel edge detection along with pixel enhancement on the PYNQ-Z2 FPGA has been well tested to figure out its performance, accuracy, and applicability in medical imaging applications. Experiments were carried out on a collection of X-ray images with a general focus of bone fracture detection and structure analysis. Comparisons occurred compared to the conventional software-based Sobel edge detection implementations to highlight the benefits in using the FPGA based processing. One of the interesting outcomes observed through the experiments was the drastic improvement in the processing time. With CPU implementations requiring several seconds to process high resolution X-ray images, the hardware implementation developed in the FPGA was an opportunity to achieve better edge detection results in real time. This improvement was achieved by taking the parallel nature of the FPGA architecture and allowing multiple convolution operations to be performed at the same time. Compared with the execution of the same feature with sequential CPU, the FPGA implementation showed a significant increase in the speed and efficiency. A further comment was in connection with the clarity of the resultant images using edge detection. The Sobel operator was effective in enhancement of structural contours, and when combined with the hardware pixel enhancement phase, the final images displayed enhanced contrast of support for diagnostically important features. Bone fractures, small cracks and tiny structural

pathology that were less visible in the first X-ray images were much more visible in the enhanced outputs. This addition helps physicians make better decisions without needing to resort to costly post processing systems. The hardware design was indicative of a good distribution of logic elements, LUTs and DSP slices (Fig.2).

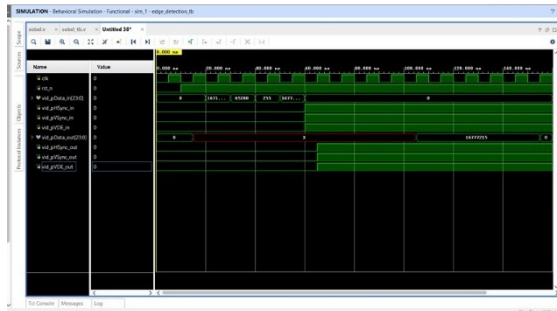


Fig 2: Output Simulation

Analysis was also performed on parameters related to resource consumption on the PYNQ-Z2 FPGA. The power consumption was also considered and found to be significantly lesser than GPU-based implementations and therefore, the proposed system was also of great importance for portable devices used in the healthcare domain. The balance of performance and resource effectiveness whom how FPGA based design is not just efficient but economical in the application space of healthcare. Latency analysis results that our work has a processing latency around few milliseconds. This result ensured real-time capability, a critical requirement in the case of ER or transportable diagnostic kits application. The software implementations added considerable latency that might hinder on-time diagnosis. The system based on an FPGA is best suited to applications at the point of care because the system is able to achieve nearly instantaneous results and reliability is paramount. Comparisons of accuracy were made by superimposing the results produced by the FPGA on edge maps with ground-truth information or software-generated high-quality results. The results from the FPGA were found to be almost undistinguishable in terms of positioning of the edges and intensity display. This result supported the fact that the hardware implementation did not compromise diagnostic accuracy but produced results faster. Additionally, subjective evaluations by clinicians indicated from the enhanced edge outputs that the enhanced visualization of fractures and bone structure occurred compared to conventional images. Scalability in design was also investigated. Through the modularity of FPGA reconfigurability and coding in Verilog HDL, the system is expandable to support other operators of edge detection or to be integrated with higher-level diagnostic algorithms. The design is also scalable with little change to the whole pipeline, which means it is possible to have design flexibility in the future.

The overall analysis presented that the Sobel edge detection system based on an FPGA, extended through pixel capabilities was able to perform well in real time while considerably improving the quality, resource usage and portability of the diagnostic analysis. These characteristics make it an effective and advanced innovation applicable in

modern healthcare especially in the area's with limited installations of advanced diagnostic facilities

Table 1

COMPARISON OF FPGA BASED EDGE DETECTION

Metrics	Abdullah et al. (2024) [22]	Ravichandran et al. (2025) [23]	Our Work (2025)
Platform	Spartan3E XC3S1600	DE1-SoC (Cyclone V)	PYNQ-Z2
LUTs	371 slices	473 slices	208 slices
Latency (ms)	11.76	3.3	4.8
Throughput (FPS)	85	303	208
Edge Detection Method	Sobel, Prewitt, Roberts	Enhanced Sobel(8-directional)	Sobel with Pixel Enhancement

V. CONCLUSION

This work is a report on a hardware based design of Sobel edge detection and pixel enhancement combined on the PYNQ-Z2 FPGA board specifically designed for medical imaging. For example, the idea was to get around some of the disadvantages of traditional CPU and GPU-based approaches, which are typically not real-time and tend to be relatively expensive and require a lot of power. Through the exploiting the parallel processing characteristic of FPGAs the system designed here effectively attained real-time enhancement of the input image amenable to diagnostic application in the healthcare domain. Simulation results are obtained in terms of low latency and high accuracy in edge detection in grayscale X-ray images. The pixel enhancement level further improved the clarity of some important features such as bone fractures and structural defects to assist clinicians in the reliable diagnosis. Analysis of resource utilization showed the design to achieve effective hardware utilization while allowing room to incorporate other modules or enhanced imaging capabilities in the future. The power measurement also verified suitability in portable and low-powered devices in medicine. The innovation of the proposed work lies in the combination of edge detection with real-time enhancement within the hardware itself in order to avoid post-processing from other systems. This offers the solution a wide applicability level at the point-of-care diagnostics level for the case of low-resource or rural health centers without a connection with advanced imaging systems. The edge detection and real-time enhancement system using a Sobel approach based on an FPGA offers a transportable, economic, and effective system for medical imaging support.

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